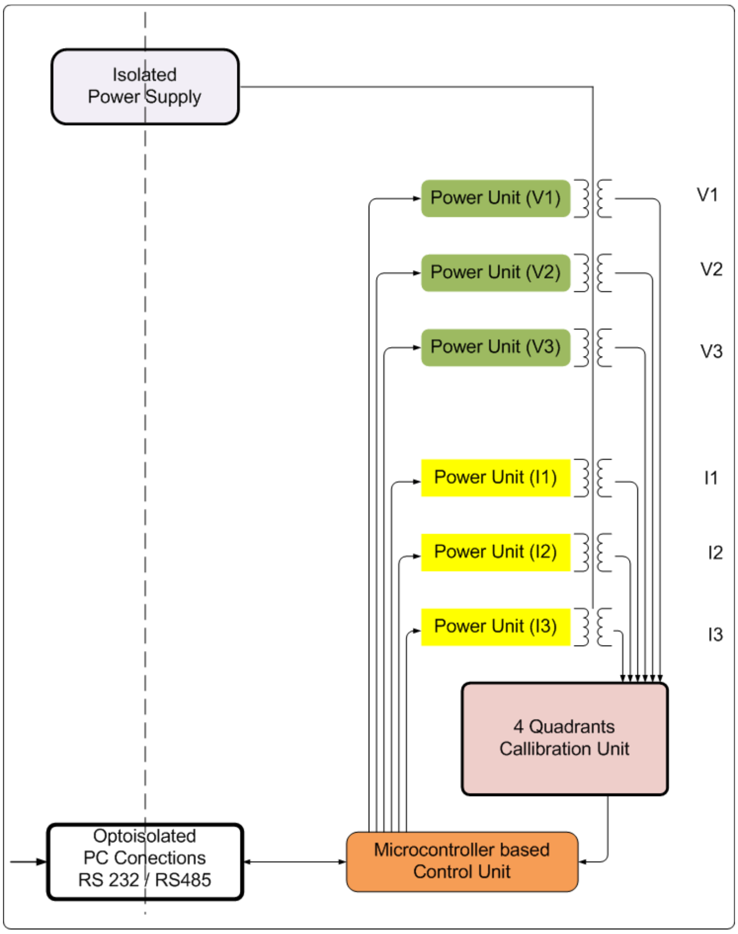
GPTR (Generador Programable de red Trifásica)

Programmable 3 Phase Grid Generator and 4 quadrants Load/Generator Simulator

Description:

This instrument creates a 3 phase Grid controllable in every aspect (voltages, frequency, angle between phases) and a set of three current generators that mimics the output of current transformers in control panels of Power installations. The combinations are 4 quadrants so it can emulate both load and generation.

Grid (Voltage generators):
Three sinusoidal synchronous generators (V1,V2,V3). The voltage at each generator can independently be set from *0 to 420 Vrms phase to neutral (690 between phases) with 1 V rms resolution)*. The angle difference fo Phases 2 and 3 can be independently be changed respect to to Phase 1 (V1 is used as angle reference for all other generators), between 0 to 360 °, in 1° steps.
Current generators (to emulate any load):
Three sinusoidal generators (I1,I2,I3), synchronous with the voltage generators, able to generate currents in two ranges: 1) from *0 to 1 A rms* or 2) from *0 to 5 A rms*. The amplitude and phase angle (respect to V1) of each generator can independently be changed: amplitude in 1 mA steps, angles between 0 to 360° with 1° resolution.
Both sets of generators are synchronous and have galvanic isolation among them



Characteristics:
V1,V2,V3: 0 to 420 V rms in 1 v steps. Current limited to 12 mA, independently on each channel.
I1,I2,I3: 0 to 1 A rms or 0 to 5 A rms in 1 A steps.
Phase angles: 0 to 360° respect to V1, in 1° steps.
Frequency: 39 to 65 Hz in 1 Hz steps.
Comunications: RS485 half duplex, 9600 bauds.
Commands: For setting and Reading each of the parameters. In ASCII Format.
Control software: 1) A control program with a ready to use user interface (see next figure); 2) A DLL usable from Visual C, VisualBasic, Excel, etc.; 3) An Excel WorkBook that uses the DDL.
Calibration: Includes a calibrated Grid Analyzer that is used by the Calibration command to adjust any changes in the supply or the load.

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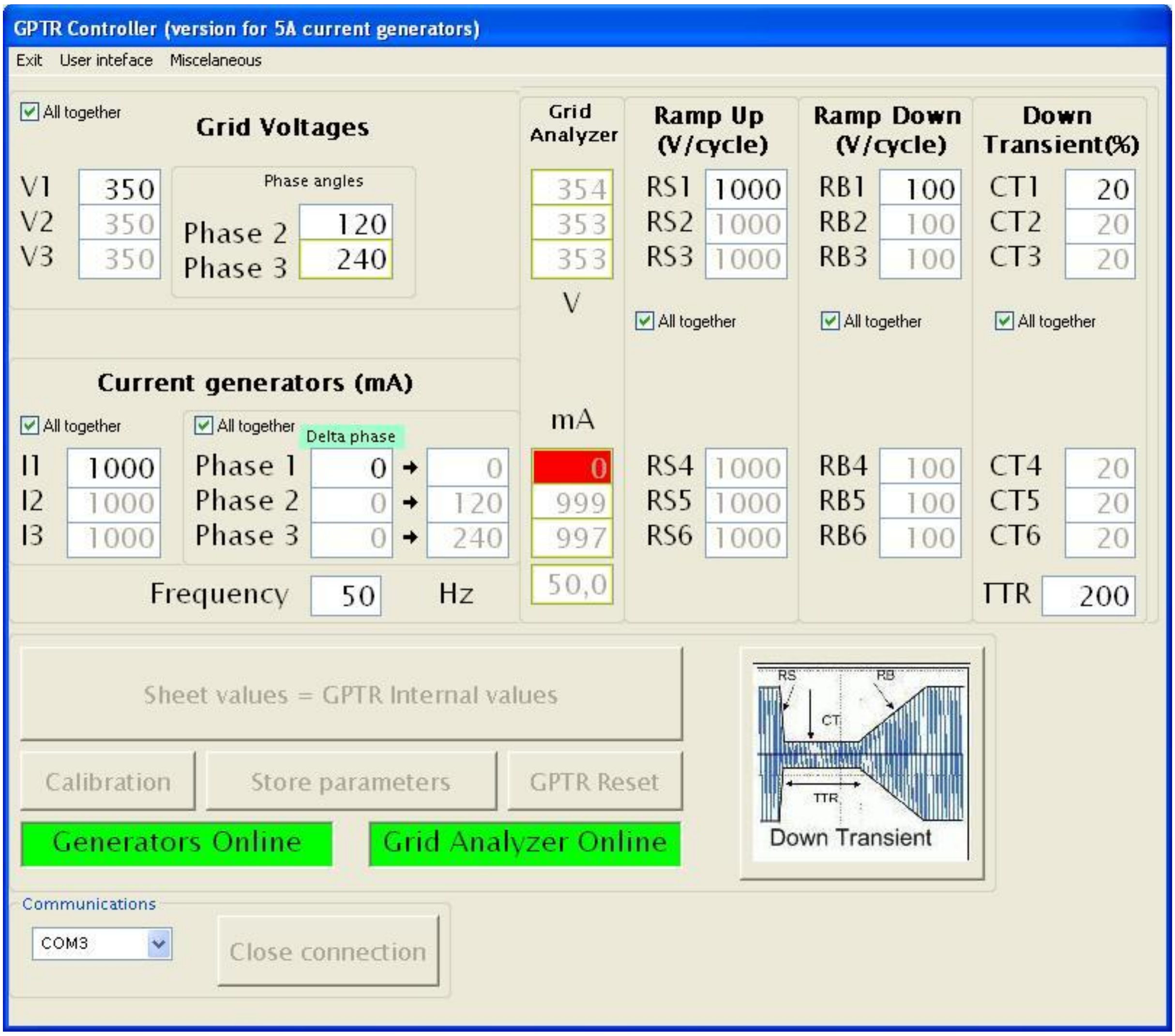

Control Software:

The GPTR is supplied with a program running on Windows XP, Windows 2000, with the User Interface shown in the picture above, designed for easy of access to every parameter inside the GPTR.

A DLL is also supplied that eases the communications and control of the internal parameters of the GPTR from standard programming languages (VisualBasic, VisualC++, Delphi,CBuilder, etc).

An example of use of this DLL is supplied as a Excel WorkBook.

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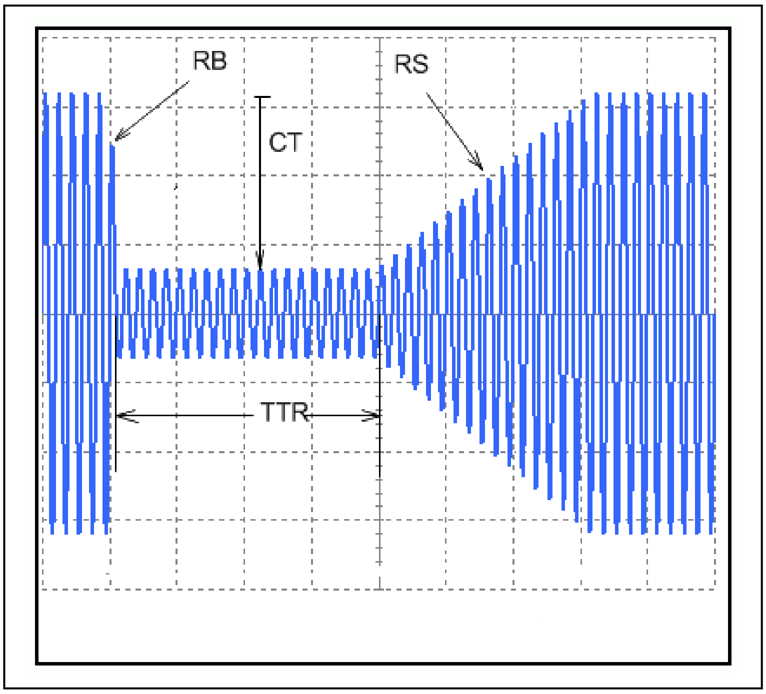
HT option:

This option allows the definition of commands sequences that control the creation of voltage and/or current profiles and to execute them in a single command. The parameters that control the form of the generated profile are shown in the next figure and can be controlled independently on each cannel (voltage or current).



Aplications:

GPTR is appropriate for applications where a controllable and operationally saved 3 phase grid o single phase grid is required, such as:

- For Simulation of Grids in Electrical Test Benches.
- Test of algorithms embedded in PLCs.
- Tests in operational conditions of Service Panels in a save way.
- Simulators.
- Test of Control Units for Cogeneration, wind turbines, and similar plants.
- Calibration of Grid Analyzers

Block diagram and typical connection to a Unit Under Test